

Tikhonov Regularization

Synopsis

The method of Tikhonov regularization for stabilizing inverse problem solutions is introduced and illustrated with examples. Zeroth-order Tikhonov regularization is explored, including its resolution, bias, and uncertainty properties. The concepts of filter factors (which control the contribution of singular values and their corresponding singular vectors to the solution) and the discrepancy and L-curve criteria (strategies for selecting the regularization parameter) are presented. Higher-order Tikhonov regularization techniques and their computation by the generalized singular value decomposition (GSVD) and truncated GSVD are discussed.

4.1. SELECTING GOOD SOLUTIONS TO ILL-POSED PROBLEMS

We saw in Chapter 3 that, given the SVD of $\mathbf{G}$ (3.1), we can express a generalized inverse solution by a series (3.81)

$$\mathbf{m}_{\dagger} = \mathbf{V}_p \mathbf{S}_p^{-1} \mathbf{U}_p^T \mathbf{d} = \sum_{i=1}^p \frac{\mathbf{U}_{:,i}^T \mathbf{d}}{s_i} \mathbf{V}_{:,i}. \quad (4.1)$$

We also saw that the generalized inverse solution can become extremely unstable when one or more of the singular values, s_i , is small. One approach for dealing with this difficulty, the truncated singular value decomposition (TSVD), was to truncate the series to remove singular vectors, $\mathbf{V}_{:,i}$, associated with smaller singular values, s_i . This stabilized, or regularized, the solution in the sense that it made the result less sensitive to data noise. The cost of this stabilization approach is that the regularized solution had reduced resolution and was no longer unbiased.

In this chapter we will introduce and discuss Tikhonov regularization, a very widely applied and easily implemented technique for regularizing discrete ill-posed problems. We will show a series formula for the Tikhonov solution that is a modified version of the generalized inverse series (3.81). The Tikhonov series solution has coefficients that are functions of the regularization parameter controlling the degree of regularization and which give greater weight to model elements associated with larger singular values.

For a general linear least squares problem there may be infinitely many least squares solutions. If we consider that the data contain noise, and that there is no point in fitting such noise exactly, it becomes evident that there can be many solutions that can adequately fit the data in the sense that $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$ is small enough.

In zeroth-order Tikhonov regularization, we consider all solutions with $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2 \leq \delta$, and select the one that minimizes the norm of $\mathbf{m}$

$$\begin{aligned} \min & \|\mathbf{m}\|_2 \\ & \|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2 \leq \delta. \end{aligned} \quad (4.2)$$

The fundamental motivation for minimizing the norm of $\mathbf{m}$ is to arrive at a solution that contains just sufficient feature complexity, as quantified by a norm measure, to adequately fit the data. Note that as δ increases, the set of feasible models expands, and the minimum value of $\|\mathbf{m}\|_2$ decreases. In other words, as we allow a poorer fit to the data, a smaller norm model will suffice to fit the data. We can thus trace out a curve of minimum values of $\|\mathbf{m}\|_2$ versus δ (Figure 4.1). It is also possible to trace out this curve by considering problems of the form

$$\begin{aligned} \min & \|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2 \\ & \|\mathbf{m}\|_2 \leq \epsilon. \end{aligned} \quad (4.3)$$

As ϵ decreases, the set of feasible solutions becomes smaller, and the minimum value of $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$ increases. Again, as we adjust ϵ we trace out the curve of optimal values of $\|\mathbf{m}\|_2$ and $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$ (Figure 4.2).

A third option is to consider the **damped least squares** problem

$$\min \|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{m}\|_2^2, \quad (4.4)$$

which arises when we apply the method of Lagrange multipliers to (4.2), where α is a **regularization parameter**. It can be shown that for appropriate choices of δ , ϵ , and α , the three problems (4.2), (4.3), and (4.4) yield the same solution [65]. We will

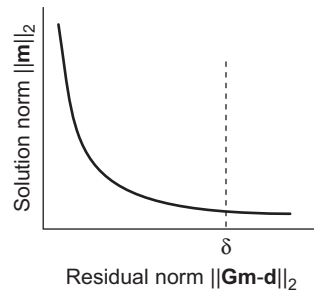


Figure 4.1 A particular misfit norm, δ , and its position on the trade-off curve between residual misfit, $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$, and model norm, $\|\mathbf{m}\|_2$.

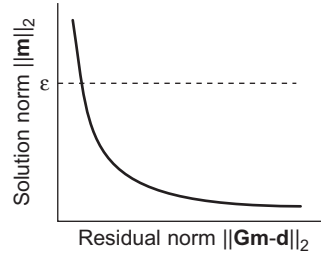


Figure 4.2 A particular model norm, ϵ , and its position on the trade-off curve between residual misfit, $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$, and model norm, $\|\mathbf{m}\|_2$.

concentrate on solving the damped least squares form of the problem (4.4). Solutions to (4.2) and (4.3) can be obtained using (4.4) by adjusting the regularization parameter α until the constraints are just satisfied.

When plotted on a log-log scale, the curve of optimal values of $\|\mathbf{m}\|_2$ versus $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$ often takes on a characteristic “L” shape in linear problems. This happens because $\|\mathbf{m}\|_2$ is a strictly decreasing function of α and $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$ is a strictly increasing function of α . The sharpness of the “corner” varies from problem to problem, but it is frequently well-defined. For this reason, the curve is called an **L-curve** [63]. In addition to the discrepancy principle, another popular criterion for picking the value of α is the **L-curve criterion**, in which the value of α that gives the solution closest to the corner of the L-curve is selected.

4.2. SVD IMPLEMENTATION OF TIKHONOV REGULARIZATION

The damped least squares problem (4.4) is equivalent to the ordinary least squares problem obtained by augmenting the least squares problem for $\mathbf{G}\mathbf{m} = \mathbf{d}$ in the following manner:

$$\min \left\| \begin{bmatrix} \mathbf{G} \\ \alpha \mathbf{I} \end{bmatrix} \mathbf{m} - \begin{bmatrix} \mathbf{d} \\ \mathbf{0} \end{bmatrix} \right\|_2^2. \quad (4.5)$$

As long as α is nonzero, the last n rows of the augmented matrix in (4.5) are obviously linearly independent. Equation (4.5) is thus a full rank least squares problem that can be solved by the method of normal equations, that is,

$$\begin{bmatrix} \mathbf{G}^T & \alpha \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{G} \\ \alpha \mathbf{I} \end{bmatrix} \mathbf{m} = \begin{bmatrix} \mathbf{G}^T & \alpha \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{d} \\ \mathbf{0} \end{bmatrix}. \quad (4.6)$$

Equation (4.6) simplifies to

$$(\mathbf{G}^T \mathbf{G} + \alpha^2 \mathbf{I}) \mathbf{m} = \mathbf{G}^T \mathbf{d}, \quad (4.7)$$

which is the set of constraint equations for a **zeroth-order Tikhonov regularization** solution of $\mathbf{G}\mathbf{m} = \mathbf{d}$.

Employing the SVD of $\mathbf{G}$, (4.7) can be written as

$$(\mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{U}\mathbf{S}\mathbf{V}^T + \alpha^2\mathbf{I})\mathbf{m} = \mathbf{G}^T\mathbf{d} \quad (4.8)$$

$$(\mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T + \alpha^2\mathbf{I})\mathbf{m} = \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{d}. \quad (4.9)$$

A simple substitution diagonalizes this system of equations and makes it straightforward to write out the solution. Let $\mathbf{x} = \mathbf{V}^T\mathbf{m}$ and $\mathbf{m} = \mathbf{V}\mathbf{x}$. Since $\mathbf{V}\mathbf{V}^T = \mathbf{I}$, we can write (4.9) as

$$(\mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T + \alpha^2\mathbf{V}\mathbf{V}^T)\mathbf{m} = \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{d}. \quad (4.10)$$

$$\mathbf{V}(\mathbf{S}^T\mathbf{S} + \alpha^2\mathbf{I})\mathbf{V}^T\mathbf{m} = \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{d}. \quad (4.11)$$

$$(\mathbf{S}^T\mathbf{S} + \alpha^2\mathbf{I})\mathbf{x} = \mathbf{S}^T\mathbf{U}^T\mathbf{d}. \quad (4.12)$$

The matrix on the left-hand side of this equation is diagonal, so it is trivial to solve the system of equations,

$$x_i = \frac{s_i \mathbf{U}_{:,i}^T \mathbf{d}}{s_i^2 + \alpha^2}. \quad (4.13)$$

Since $\mathbf{m} = \mathbf{V}\mathbf{x}$, we obtain the solution

$$\mathbf{m}_\alpha = \sum_{i=1}^k \frac{s_i \mathbf{U}_{:,i}^T \mathbf{d}}{s_i^2 + \alpha^2} \mathbf{V}_{:,i} \quad (4.14)$$

where $k = \min(m, n)$ so that all of the nonzero singular values and vectors are included.

To relate this formula to (4.1), we can rewrite it slightly as

$$\mathbf{m}_\alpha = \sum_{i=1}^k \frac{s_i^2}{s_i^2 + \alpha^2} \frac{\mathbf{U}_{:,i}^T \mathbf{d}}{s_i} \mathbf{V}_{:,i} \quad (4.15)$$

or

$$\mathbf{m}_\alpha = \sum_{i=1}^k f_i \frac{\mathbf{U}_{:,i}^T \mathbf{d}}{s_i} \mathbf{V}_{:,i}. \quad (4.16)$$

Here, the **filter factors**

$$f_i = \frac{s_i^2}{s_i^2 + \alpha^2} \quad (4.17)$$

control the contribution to the sum from different terms. For $s_i \gg \alpha$, $f_i \approx 1$, and for $s_i \ll \alpha$, $f_i \approx 0$. For singular values between these two extremes, as the s_i decrease,

the f_i produce a monotonically decreasing contribution of corresponding model space vectors, $\mathbf{V}_{:,i}$.

A related method called the **damped SVD method** [65] uses the filter factors

$$\hat{f}_i = \frac{s_i}{s_i + \alpha}. \quad (4.18)$$

This has a similar effect to using (4.17), but transitions more slowly with the index i .

Example 4.1

We revisit the severely ill-posed Shaw problem, which was previously introduced in Examples 1.6 and 1.10, and was analyzed using the SVD in Example 3.3. The true model in this synthetic example is a spike of unit amplitude in the 10th model element, and independent $N(0, (10^{-6})^2)$ noise has been added to the data vector. We begin by computing the L-curve at 1000 points and finding its corner by estimating the point of maximum curvature. Figure 4.3 shows the L-curve. The corner of the curve corresponds to $\alpha \approx 6.40 \times 10^{-6}$.

Next, we compute the Tikhonov regularization solution corresponding to this value of α . This solution is shown in Figure 4.4. Note that this solution is much better than the wild solution obtained by the TSVD with $p' = 18$ (Figure 3.24).

Alternatively, we can use the discrepancy principle to find an appropriate α to obtain a Tikhonov regularized solution. Because independent $N(0, (1 \times 10^{-6})^2)$ noise was added to these $m = 20$ data points, we search for a solution for which the square of the

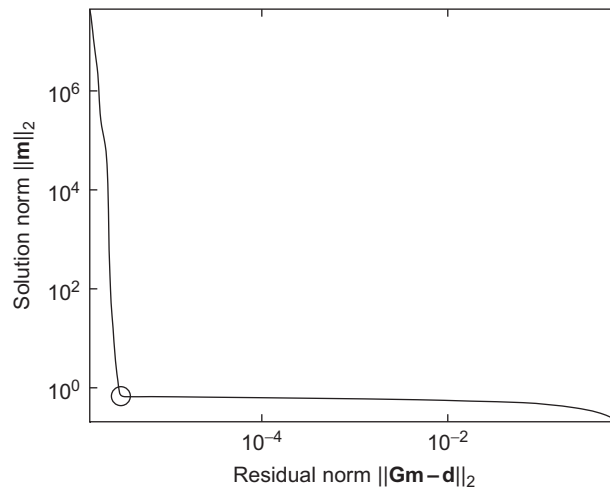


Figure 4.3 Zeroth-order Tikhonov regularization L-curve for the Shaw problem, with corner estimated using maximum functional curvature.

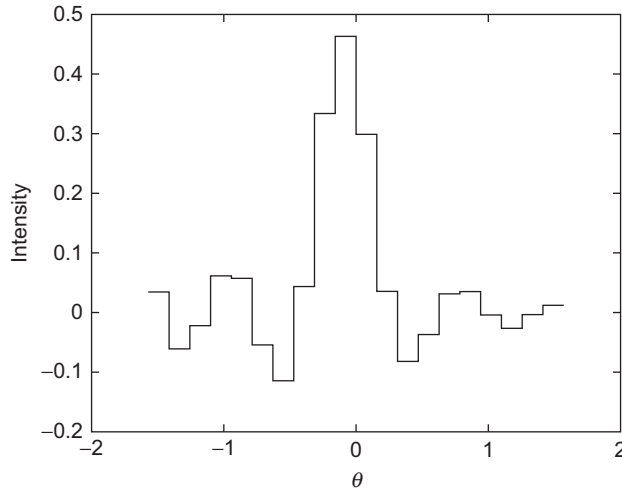


Figure 4.4 Recovery of the spike model with noise, zeroth-order Tikhonov solution ($\alpha = 6.40 \times 10^{-6}$ determined from the L-curve corner).

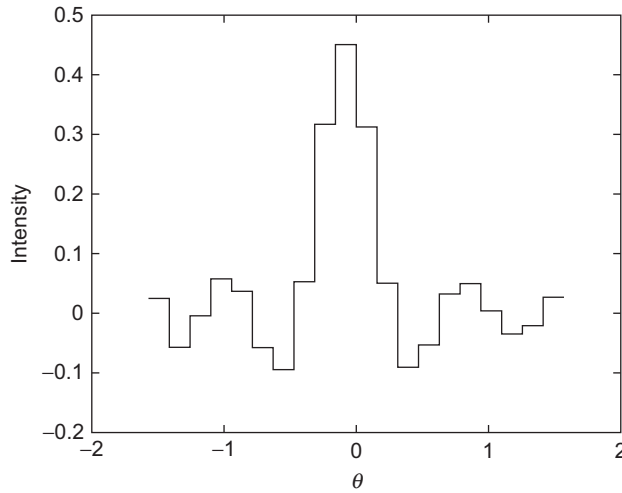


Figure 4.5 Recovery of the spike model with noise, zeroth-order Tikhonov solution ($\alpha = 4.29 \times 10^{-5}$ determined from the discrepancy principle).

norm of the residuals is 20×10^{-12} , which corresponds to a residual norm $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$ of $\sqrt{20} \times 10^{-6} \approx 4.47 \times 10^{-6}$.

The discrepancy principle results in a somewhat larger value of the regularization parameter, $\alpha = 4.29 \times 10^{-5}$, than that obtained using the L-curve technique above. The corresponding solution, shown in Figure 4.5, thus has a smaller model norm, but the two models are quite close.

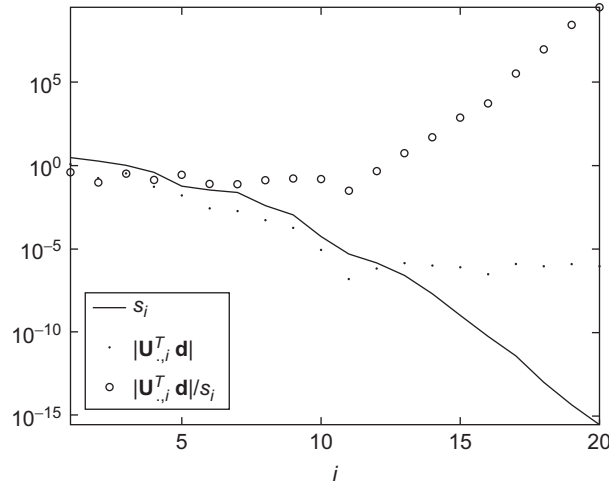


Figure 4.6 Picard plot for the Shaw problem of [Example 4.1](#).

It is interesting to note that the norm of the residual of the true (spike) model in this particular case, 3.86×10^{-6} , is actually slightly smaller than the tolerance that we specified via the discrepancy principle (4.47×10^{-6}). The discrepancy principle method did not recover the original spike model because the spike model has a norm of 1, while the solution obtained by the discrepancy principle has a norm of only 0.67.

Plotting the singular values s_i , the values of $|\mathbf{U}_{:,i}^T \mathbf{d}|$, and the ratios $|\mathbf{U}_{:,i}^T \mathbf{d}|/s_i$ allows us to examine the discrete Picard condition for this problem ([Figure 4.6](#)). $|\mathbf{U}_{:,i}^T \mathbf{d}|$ reaches a noise floor of about 1×10^{-6} after $i = 11$. The singular values continue to decay. As a consequence, the ratios increase rapidly. It is clear from this plot that we cannot expect to obtain useful information from the singular values beyond $i = 11$. The 11th singular value is $\approx 5.1 \times 10^{-6}$, which is comparable to the values of α in [Figures 4.4](#) and [4.5](#).

4.3. RESOLUTION, BIAS, AND UNCERTAINTY IN THE TIKHONOV SOLUTION

As in our earlier TSVD approach, we can compute a model resolution matrix for the Tikhonov regularization method. Using [equation \(4.7\)](#) and the SVD, the solution can be written as

$$\begin{aligned}
 \mathbf{m}_\alpha &= (\mathbf{G}^T \mathbf{G} + \alpha^2 \mathbf{I})^{-1} \mathbf{G}^T \mathbf{d} \\
 &= \mathbf{G}^\dagger \mathbf{d} \\
 &= \mathbf{VFS}^\dagger \mathbf{U}^T \mathbf{d},
 \end{aligned} \tag{4.19}$$

where $\mathbf{F}$ is an n by n diagonal matrix with diagonal elements given by the filter factors f_i of (4.17), and $\mathbf{S}^\dagger$ is the pseudoinverse of $\mathbf{S}$. $\mathbf{G}^\sharp$ is a generalized inverse matrix that can be used to construct model and data resolution matrices as was done for the SVD solution in (3.63) and (3.77). The resolution matrices are

$$\mathbf{R}_m = \mathbf{G}^\sharp \mathbf{G} = \mathbf{V} \mathbf{F} \mathbf{V}^T \quad (4.20)$$

and

$$\mathbf{R}_d = \mathbf{G} \mathbf{G}^\sharp = \mathbf{U} \mathbf{F} \mathbf{U}^T. \quad (4.21)$$

Note that the resolution matrices are dependent on the particular value of α (4.19).

Example 4.2

In Example 4.1, with $\alpha = 4.29 \times 10^{-5}$ as selected using the discrepancy principle, the model resolution matrix for zeroth-order Tikhonov regularization has the following diagonal elements:

$$\begin{aligned} \text{diag}(\mathbf{R}_m) \approx & [0.91, 0.49, 0.45, 0.39, 0.42, 0.41, 0.43, 0.44, 0.44, \dots \\ & 0.45, 0.45, 0.44, 0.44, 0.43, 0.41, 0.42, 0.39, 0.45, 0.49, 0.91]^T \end{aligned} \quad (4.22)$$

indicating that most model parameters are not well resolved. Figure 4.7 displays the effect of this limited resolution by applying $\mathbf{R}_m$ to the (true) spike model (3.63) or, equivalently, inverting noise-free spike model data using (4.6) for the regularization parameter

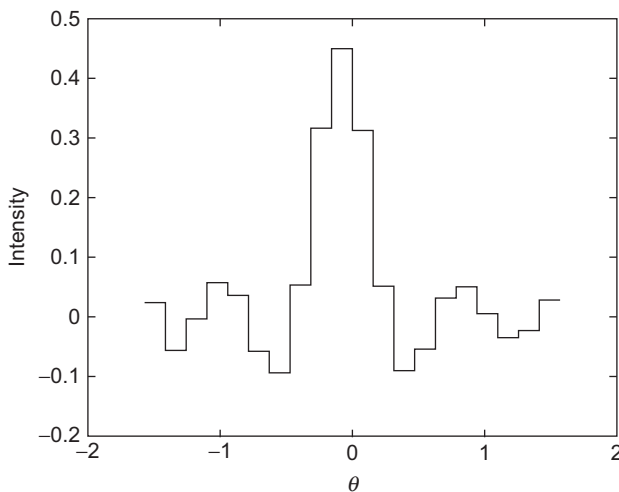


Figure 4.7 Resolution test using the spike model ($\alpha = 4.29 \times 10^{-5}$ determined from the discrepancy principle). Note that this model is nearly equivalent to that shown in Figure 4.5.



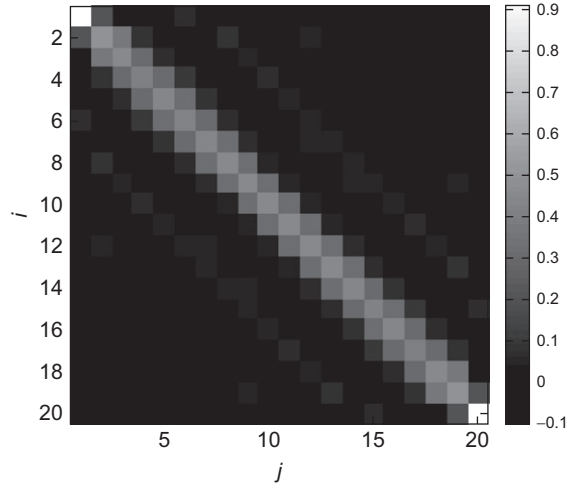


Figure 4.8 Resolution matrix for the Shaw problem, $\alpha = 4.29 \times 10^{-5}$. Note that the noise-free inversion of a spike model e_i will produce the i th column or row of $\mathbf{R}_m$ (e.g., Figure 4.7 is a plot of the 10th column/row).

value estimated using the discrepancy principle. Note that the result of limited resolution is that the true model “leaks” or “smears” into adjacent model parameters and is reduced in its maximum amplitude in the recovered model. In this example, the noise-free spike model recovery obtained in this resolution test is nearly identical to the recovery from the noisy spike model data using zeroth-order Tikhonov regularization (Figure 4.5), indicating that noise has only a very slight effect on model recovery accuracy. Thus, the differences between the true and recovered models here are essentially entirely due to the regularization that was necessary to stabilize the solution, rather than from noise propagation from data to model (see Example 4.3). Figure 4.7 displays just a single row or column from the (symmetric) $\mathbf{R}_m$, but effects of limited resolution can be examined more comprehensively by imaging the entire resolution matrix (Figure 4.8).

As in Chapter 2, we can compute a covariance matrix for the estimated model parameters using (B.65). Since

$$\mathbf{m}_\alpha = \mathbf{G}^\# \mathbf{d}, \quad (4.23)$$

the model covariance is

$$\text{Cov}(\mathbf{m}_\alpha) = \mathbf{G}^\# \text{Cov}(\mathbf{d}) (\mathbf{G}^\#)^T. \quad (4.24)$$

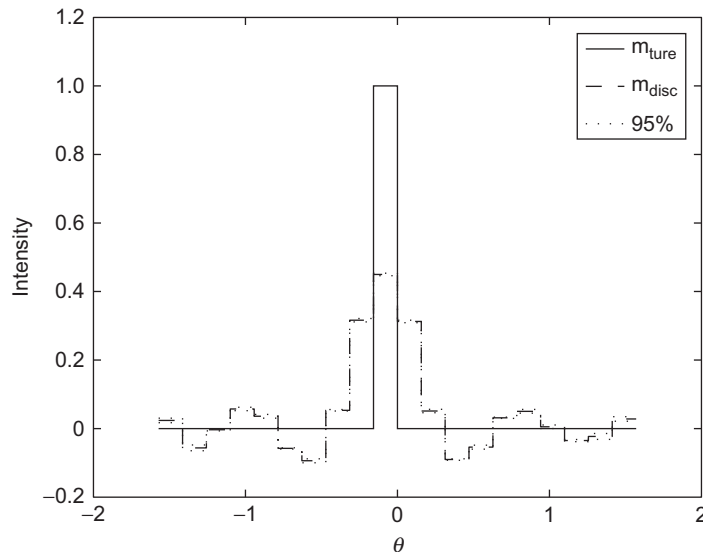


Figure 4.9 Tikhonov solution and confidence intervals for the Shaw problem, estimated using (4.24), where the true model is a spike and the data noise is independent and $N(0, (1 \times 10^{-6})^2)$. The regularization parameter $\alpha = 4.29 \times 10^{-5}$ was chosen using the discrepancy principle. The confidence interval is barely visible at this scale because inaccuracy in the model recovery is dominated by limited resolution.

Note that, as with the TSVD solution of Chapter 3, the Tikhonov regularized solution will generally be biased, and differences between the regularized solution values and the true model may actually be much larger than the confidence intervals obtained from the covariance matrix of the model parameters. Rather, the confidence intervals reflect the difference between $\mathbf{m}_\alpha$ and $\mathbf{R}_m \mathbf{m}_{\text{true}}$. See Figure 4.9.

Example 4.3

Recall our earlier example of the Shaw problem with the true spike model. Figure 4.9 shows the true model, the solution obtained using $\alpha = 4.29 \times 10^{-5}$ chosen using the discrepancy principle, and 95% confidence intervals for the estimated parameters. Note that the confidence intervals are extremely tight, and that very few of the true model parameters are included within the confidence intervals. In this case, the regularization bias, which is *not* estimated by the covariance matrix, is far larger than the propagated data uncertainty. In other words, the inaccuracy in model recovery in this case is dominated by limited resolution rather than by noise propagation from data to model. The solution shown in Figure 4.9 is essentially identical to the product of $\mathbf{R}_m$ and $\mathbf{m}_{\text{true}}$ shown in Figure 4.7 or the 10th column of the resolution matrix of Figure 4.8.



4.4. HIGHER-ORDER TIKHONOV REGULARIZATION

So far in our discussions of Tikhonov regularization we have minimized an objective function involving $\|\mathbf{m}\|_2$. In many situations, we would prefer to obtain a solution that minimizes some other measure of $\mathbf{m}$, expressed as $\mathbf{L}\mathbf{m}$, such as the norm of the first or second derivative of $\mathbf{m}$, reflecting a preference for a “flat” or “smooth” model. We then solve the regularized least squares problem

$$\min \|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{L}\mathbf{m}\|_2^2. \quad (4.25)$$

As with (4.5), this can be rewritten as a standard linear least squares problem:

$$\min \left\| \begin{bmatrix} \mathbf{G} \\ \alpha\mathbf{L} \end{bmatrix} \mathbf{m} - \begin{bmatrix} \mathbf{d} \\ \mathbf{0} \end{bmatrix} \right\|_2^2. \quad (4.26)$$

For example, if we have discretized our problem using simple collocation and our model is one-dimensional, then we can approximate, to a multiplicative constant, the first derivative of the model by a finite-difference operation $\mathbf{L}_1\mathbf{m}$, where

$$\mathbf{L}_1 = \begin{bmatrix} -1 & 1 & & & & \\ & -1 & 1 & & & \\ & & \ddots & \ddots & & \\ & & & -1 & 1 & \\ & & & & -1 & 1 \end{bmatrix}. \quad (4.27)$$

The matrices that are used to differentiate $\mathbf{m}$ for the purposes of regularization are commonly referred to as **roughening matrices**. In (4.27), $\mathbf{L}_1\mathbf{m}$ is a finite-difference approximation that is proportional to the first derivative of $\mathbf{m}$. By penalizing $\|\mathbf{L}_1\mathbf{m}\|_2$, we will favor solutions that are relatively flat (i.e., in the limit, constant). Note that $\|\mathbf{L}_1\mathbf{m}\|_2$ is a seminorm because it is zero for any constant model, not just for $\mathbf{m} = \mathbf{0}$.

In applying **second-order Tikhonov regularization** to a one-dimensional problem, we use a roughening matrix of the form

$$\mathbf{L}_2 = \begin{bmatrix} 1 & -2 & 1 & & & \\ & 1 & -2 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & -2 & 1 \\ & & & & 1 & -2 & 1 \end{bmatrix}. \quad (4.28)$$

Here, $\mathbf{L}_2\mathbf{m}$ is a finite-difference approximation that is proportional to the second derivative of $\mathbf{m}$, and the seminorm term $\|\mathbf{L}_2\mathbf{m}\|_2$ in (4.25) penalizes solutions that are rough in a squared second derivative sense. We will refer to regularization strategies of the form of (4.25) that use $\mathbf{L}_1$ and $\mathbf{L}_2$ roughening matrices as **first- and second-order Tikhonov regularization**, respectively.

If our model is higher dimensional (e.g., two or three dimensions), then the roughening matrices described here would not be appropriate. In such cases, second-order Tikhonov regularization is often implemented using a finite-difference approximation to the Laplacian operator of appropriate dimensionality (see Exercise 4.3).

To ensure that the least squares problem (4.26) has a unique solution, we will require that the matrix

$$A = \begin{bmatrix} \mathbf{G} \\ \alpha \mathbf{L} \end{bmatrix} \quad (4.29)$$

has full column rank or equivalently that $N(\mathbf{G}) \cap N(\mathbf{L}) = \{\mathbf{0}\}$.

We have already seen how to apply zeroth-order Tikhonov regularization to solve (4.25), with $\mathbf{L} = \mathbf{I}$, using the singular value decomposition (4.16). To solve and analyze higher-order Tikhonov regularization problems, we employ the **generalized singular value decomposition**, or GSVD [54, 62, 65]. The GSVD enables the solution (4.25) to be expressed as a sum of filter factors times generalized singular vectors in a manner that is analogous to the series representation of the generalized inverse solution (4.16).

Unfortunately, the definition of the GSVD and associated notation are not presently standardized. In the following, we will adhere to the conventions used by the MATLAB `gsvd` command where $\mathbf{G}$ is an m by n matrix and $\mathbf{L}$ is a p by n matrix. Although MATLAB's implementation of the GSVD can handle matrices $\mathbf{G}$ and $\mathbf{L}$ that have overlapping null spaces, we assume in the following derivation that the matrix in (4.29) has full column rank so that the solution to (4.26) will be unique. To further simplify the derivation we will also assume that $\text{rank}(\mathbf{L}) = p$. This is certainly true for the $\mathbf{L}$ matrices in (4.27) and (4.28). In general we can eliminate redundant rows from $\mathbf{L}$ to make $\mathbf{L}$ have full row rank.

Under the above assumptions there exist matrices $\mathbf{U}$, $\mathbf{V}$, $\mathbf{X}$, $\mathbf{\Lambda}$, and $\mathbf{M}$ with the following properties and relationships:

- $\mathbf{U}$ is m by m and orthogonal.
- $\mathbf{V}$ is p by p and orthogonal.
- $\mathbf{X}$ is n by n and nonsingular.
- $\mathbf{\Lambda}$ is an m by n matrix with diagonal entries that may be shifted from the main diagonal of the matrix. The diagonal entries are

$$0 \leq \Lambda_{1,k+1} \leq \Lambda_{2,k+2} \leq \cdots \leq \Lambda_{m,k+m} \leq 1 \quad (4.30)$$

where $k = 0$ when $m > n$, and $k = n - m$ when $m \leq n$.

- $\mathbf{M}$ is a p by n diagonal matrix with

$$M_{1,1} \geq M_{2,2} \geq \cdots \geq M_{p,p} \geq 0, \quad (4.31)$$

and

$$\mathbf{M}^T \mathbf{M} + \boldsymbol{\Lambda}^T \boldsymbol{\Lambda} = \mathbf{I}. \quad (4.32)$$

- The matrices $\mathbf{G}$ and $\mathbf{L}$ can be written as

$$\mathbf{G} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{X}^T \quad (4.33)$$

and

$$\mathbf{L} = \mathbf{V} \mathbf{M} \mathbf{X}^T. \quad (4.34)$$

The **generalized singular values** of $\mathbf{G}$ and $\mathbf{L}$ are

$$\gamma_i = \frac{\lambda_i}{\mu_i} \quad (4.35)$$

where

$$\lambda = \sqrt{\text{diag}(\boldsymbol{\Lambda}^T \boldsymbol{\Lambda})} \quad (4.36)$$

and

$$\mu = \sqrt{\text{diag}(\mathbf{M}^T \mathbf{M})}. \quad (4.37)$$

These definitions may seem somewhat odd, since the diagonal elements of $\boldsymbol{\Lambda}^T \boldsymbol{\Lambda}$ and $\mathbf{M}^T \mathbf{M}$ are simply squares of the diagonal elements of $\boldsymbol{\Lambda}$ and $\mathbf{M}$. The issue here is that the diagonals of $\boldsymbol{\Lambda}$ and $\mathbf{M}$ are not of the same length. The effect of these definitions is to create vectors λ and μ that are of length n , padding with zeros as needed.

Because of the ordering of λ and μ , the generalized singular values appear in ascending order with

$$\gamma_1 \leq \gamma_2 \leq \cdots \leq \gamma_n. \quad (4.38)$$

Also note that if $\mu_i = 0$, then the corresponding generalized singular value γ_i is infinite or undefined.

Letting $\mathbf{Y} = \mathbf{X}^{-T}$ (the inverse transpose), we can apply (4.33) and the orthogonality of $\mathbf{U}$ to show that

$$\mathbf{Y}^T \mathbf{G}^T \mathbf{G} \mathbf{Y} = \boldsymbol{\Lambda}^T \boldsymbol{\Lambda}. \quad (4.39)$$

Note that

$$\lambda_i = \sqrt{\mathbf{Y}_{:,i}^T \mathbf{G}^T \mathbf{G} \mathbf{Y}_{:,i}} = \|\mathbf{G} \mathbf{Y}_{:,i}\|_2. \quad (4.40)$$

Whenever λ_i is 0, this means that the corresponding column of $\mathbf{Y}$ is in $N(\mathbf{G})$. However, when λ_i is nonzero, the corresponding column of $\mathbf{Y}$ is not in $N(\mathbf{G})$. Since $\mathbf{Y}$ is nonsingular, the columns of $\mathbf{Y}$ are linearly independent. If we pick r so that $\lambda_r = 0$, but $\lambda_{r+1} \neq 0$, then r is the dimension of $N(\mathbf{G})$, and the vectors $\mathbf{Y}_{:,1}, \mathbf{Y}_{:,2}, \dots, \mathbf{Y}_{:,r}$ form

a basis for $N(\mathbf{G})$. Note that because $\text{rank}(\mathbf{G}) + \dim(N(\mathbf{G})) = n$, we have determined that $\text{rank}(\mathbf{G}) = n - r$.

Similarly, it is easy to use (4.34) and the orthogonality of $\mathbf{V}$ to show that

$$\mathbf{Y}^T \mathbf{L}^T \mathbf{L} \mathbf{Y} = \mathbf{M}^T \mathbf{M}. \quad (4.41)$$

Since $\text{rank}(\mathbf{L}) = p$, the dimension of the null space of $\mathbf{L}$ is $n - p$. Thus $\mu_{p+1}, \mu_{p+2}, \dots, \mu_n$ are 0, and the vectors $\mathbf{Y}_{\cdot, p+1}, \mathbf{Y}_{\cdot, p+2}, \dots, \mathbf{Y}_{\cdot, n}$ form a basis for $N(\mathbf{L})$. Note that the columns of $\mathbf{Y}$ are not generally orthogonal, so we have not found orthonormal bases for the null spaces of $\mathbf{G}$ and $\mathbf{L}$.

The GSVD may be computationally expensive (or intractable for very large problems) to evaluate. However, as with the SVD, once the decomposition matrices are computed, solutions to least squares problems may be calculated very easily. We begin by introducing the change of variables

$$\mathbf{Y} \mathbf{x} = \mathbf{m}. \quad (4.42)$$

With this substitution, the normal equations for (4.26) become

$$(\mathbf{G}^T \mathbf{G} + \alpha^2 \mathbf{L}^T \mathbf{L}) \mathbf{Y} \mathbf{x} = \mathbf{G}^T \mathbf{d}. \quad (4.43)$$

Using (4.33) and (4.34), and $\mathbf{Y} = \mathbf{X}^{-T}$, we have

$$(\mathbf{Y}^{-T} \mathbf{\Lambda}^T \mathbf{U}^T \mathbf{U} \mathbf{\Lambda} \mathbf{Y}^{-1} + \alpha^2 \mathbf{Y}^{-T} \mathbf{M}^T \mathbf{V}^T \mathbf{V} \mathbf{M} \mathbf{Y}^{-1}) \mathbf{Y} \mathbf{x} = \mathbf{Y}^{-T} \mathbf{\Lambda}^T \mathbf{U}^T \mathbf{d}, \quad (4.44)$$

$$(\mathbf{Y}^{-T} \mathbf{\Lambda}^T \mathbf{\Lambda} \mathbf{Y}^{-1} + \alpha^2 \mathbf{Y}^{-T} \mathbf{M}^T \mathbf{M} \mathbf{Y}^{-1}) \mathbf{Y} \mathbf{x} = \mathbf{Y}^{-T} \mathbf{\Lambda}^T \mathbf{U}^T \mathbf{d}, \quad (4.45)$$

and, finally,

$$(\mathbf{\Lambda}^T \mathbf{\Lambda} + \alpha^2 \mathbf{M}^T \mathbf{M}) \mathbf{x} = \mathbf{\Lambda}^T \mathbf{U}^T \mathbf{d}. \quad (4.46)$$

The matrix on the left-hand side of this system of equations is diagonal. Multiplying both sides by the inverse of this diagonal matrix gives the solution

$$x_i = \frac{\lambda_i \mathbf{U}_{\cdot, i+k}^T \mathbf{d}}{\lambda_i^2 + \alpha^2 \mu_i^2}, \quad (4.47)$$

where, as in (4.30), $k = 0$ when $m > n$, and $k = n - m$ when $m \leq n$. In terms of the generalized singular values, this can be written as

$$x_i = \frac{\gamma_i^2}{\gamma_i^2 + \alpha^2} \frac{\mathbf{U}_{\cdot, i+k}^T \mathbf{d}}{\lambda_i}. \quad (4.48)$$

Substituting this expression for x_i into $\mathbf{m} = \mathbf{Y} \mathbf{x}$, we obtain the summation formula

$$\mathbf{m}_{\alpha, L} = \sum_{i=1}^n \frac{\gamma_i^2}{\gamma_i^2 + \alpha^2} \frac{\mathbf{U}_{\cdot, i+k}^T \mathbf{d}}{\lambda_i} \mathbf{Y}_{\cdot, i} \quad (4.49)$$

where

$$f_i = \frac{\gamma_i^2}{\gamma_i^2 + \alpha^2} \quad (4.50)$$

are **GSVD filter factors** that are analogous to the filter factors obtained in the series expression for the zeroth-order Tikhonov regularized solution (4.17). In evaluating this sum, we sometimes encounter situations in which γ_i is infinite. In those cases, the filter factor f_i should be set to 1. Similarly, there are situations in which $\lambda_i = 0$ and $\gamma_i = 0$, producing an expression of the form $0^2/0$ in the sum. These terms should be treated as 0.

When $\mathbf{G}$ comes from a Fredholm integral equation of the first kind, the GSVD typically has two properties that were also characteristic of the SVD. First, the m nonzero generalized singular values $\gamma_n, \gamma_{n-1}, \gamma_{n-m+1}$ from (4.35) trend toward zero without any obvious break. Second, the vectors $\mathbf{U}_{:,i}$, $\mathbf{V}_{:,i}$, $\mathbf{X}_{:,i}$, and $\mathbf{Y}_{:,i}$ tend to become rougher as γ_i decreases.

Example 4.4

We return to the vertical seismic profiling example previously discussed in Examples 1.3 and 1.9. Here, consider a 1-km deep borehole experiment discretized using $m = n = 50$ observation and model points, corresponding to sensors every 20 m, and 20 m thick, constant-slowness model layers. Figure 4.10 shows the test model that we will try to recover. A synthetic data set was generated with $N(0, (2 \times 10^{-4}\text{s})^2)$ noise added.

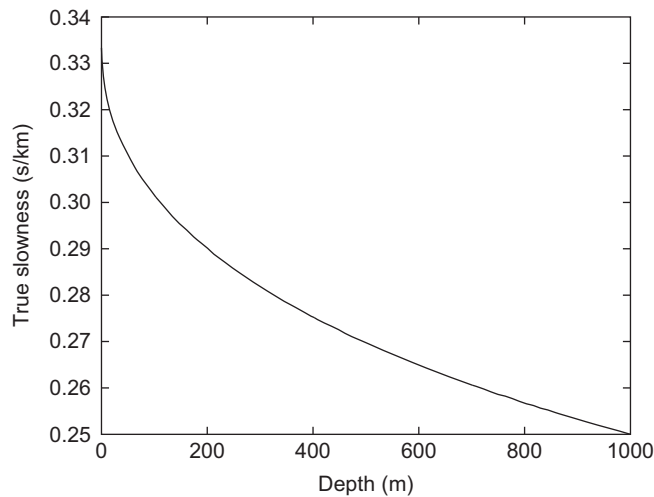


Figure 4.10 A smooth test model for the VSP problem.

The discretized system of equations $\mathbf{Gm} = \mathbf{d}$ has a small condition number (64). This happens in part because we have chosen a very coarse discretization, which effectively regularizes the problem by discretization. Another reason is that the vertical seismic profiling problem is only mildly ill-posed [45]. Figure 4.11 shows the least squares solution, together with 95% confidence intervals.

From the statistical point of view, this solution is completely acceptable. However, suppose that from other information, we believe the slowness should vary smoothly with depth. We will next apply higher-order Tikhonov regularization to obtain smooth solutions to this problem.

Figure 4.12 shows the first-order Tikhonov regularization L-curve for this problem. The L-curve has a distinct corner near $\alpha \approx 122$. Figure 4.13 shows the corresponding solution. The first-order regularized solution is much smoother than the least squares solution, and is much closer to the true solution.

Figure 4.14 shows the L-curve for second-order Tikhonov regularization, which has a corner near $\alpha \approx 1965$. Figure 4.15 shows the corresponding solution. This solution is smoother still compared to the first-order regularized solution. Both the first- and second-order solutions depart most from the true solution at shallow depths where the true slowness has the greatest slope and curvature. This happens because the first- and second-order Tikhonov regularized solutions are biased towards flatness and second-derivative smoothness, respectively.

Figure 4.16 shows filter factors corresponding to these first- and second-order solutions. Higher-order terms in (4.49) are severely downweighted in both cases, particularly

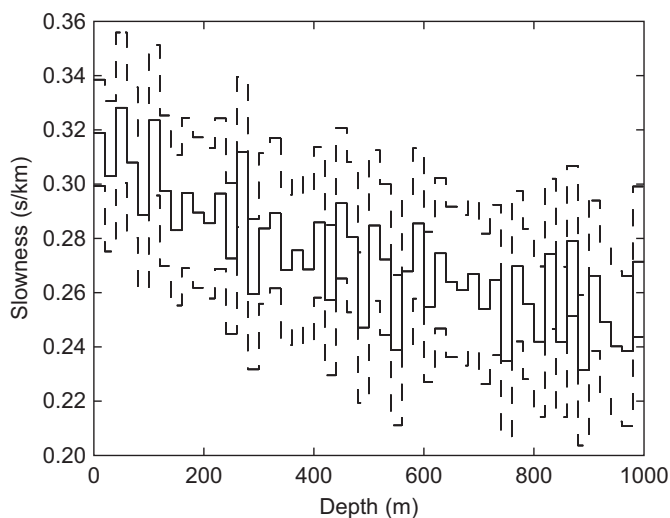


Figure 4.11 Least squares solution for the VSP problem, with 95% confidence intervals.

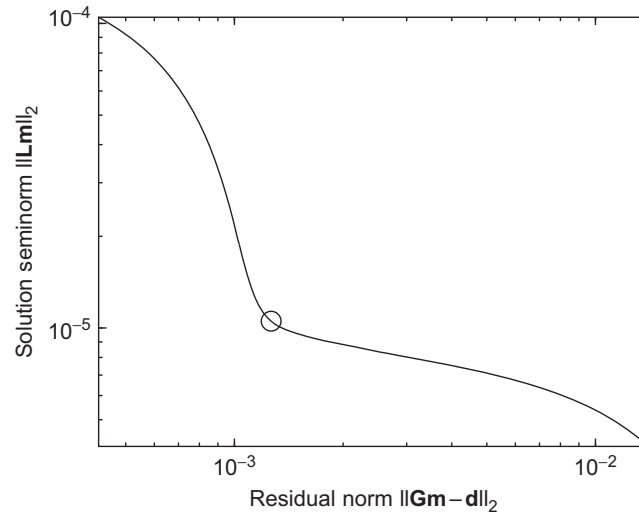


Figure 4.12 L-curve and corner for the VSP problem, first-order regularization.

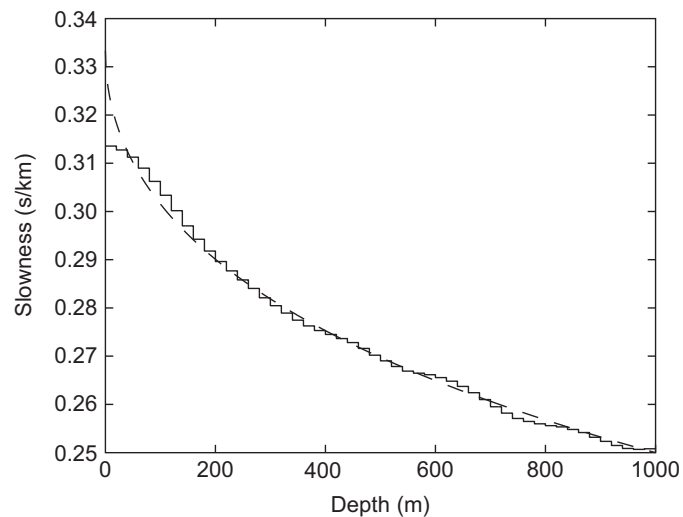


Figure 4.13 Tikhonov solution for the VSP problem, first-order regularization, $\alpha = 122$, shown in comparison with the true model (Figure 4.10).

in the second-order case. Because of the smoothness of the true model, the model seminorms can be reduced considerably through the selection of relatively large regularization parameters, α , without large data misfit increases. In this example the 2-norms

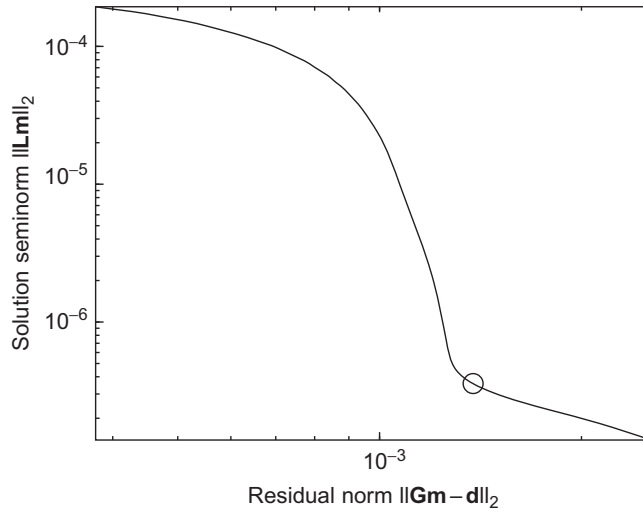


Figure 4.14 L-curve and corner for the VSP problem, second-order regularization.

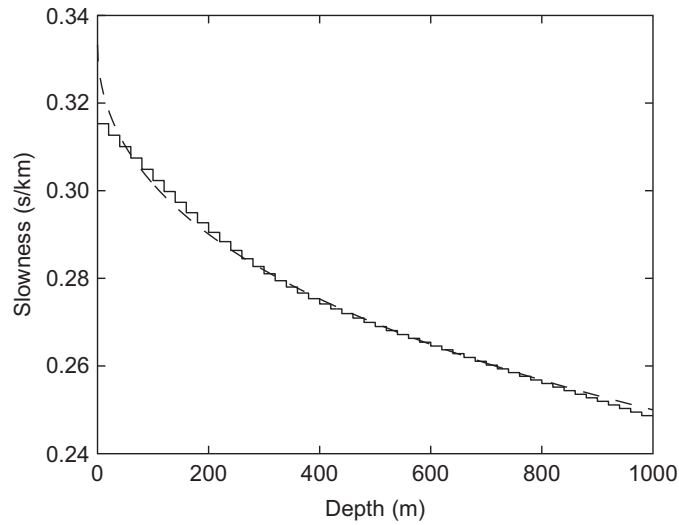


Figure 4.15 Tikhonov solution for the VSP problem, second-order regularization, $\alpha = 2341$, shown in comparison with the true model (Figure 4.10).

of the difference between the first- and second-order solutions and the true model (discretized into 50 values) are approximately 1.2×10^{-5} s/km and 1.0×10^{-5} s/km, respectively.

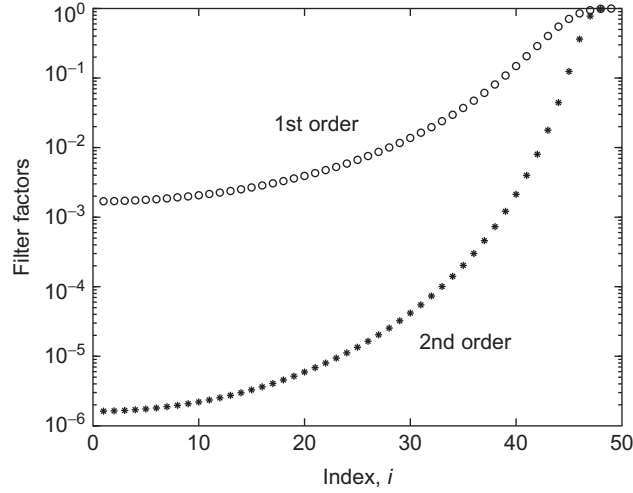


Figure 4.16 Filter factors (4.17) for optimal first- and second-order Tikhonov solutions to the VSP problem shown in Figures 4.13 and 4.15.

4.5. RESOLUTION IN HIGHER-ORDER TIKHONOV REGULARIZATION

As with zeroth-order Tikhonov regularization, we can compute a resolution matrix for higher-order Tikhonov regularization. For a particular roughening matrix $\mathbf{L}$ and value of α , the Tikhonov regularization solution can be written as

$$\mathbf{m}_{\alpha,L} = \mathbf{G}^{\sharp} \mathbf{d} \quad (4.51)$$

where

$$\mathbf{G}^{\sharp} = (\mathbf{G}^T \mathbf{G} + \alpha^2 \mathbf{L}^T \mathbf{L})^{-1} \mathbf{G}^T. \quad (4.52)$$

Using the GSVD, we can write this expression as

$$\mathbf{G}^{\sharp} = (\mathbf{X} \mathbf{\Lambda}^T \mathbf{U}^T \mathbf{U} \mathbf{\Lambda} \mathbf{X}^T + \alpha^2 (\mathbf{X} \mathbf{M}^T \mathbf{V}^T \mathbf{V} \mathbf{M} \mathbf{X}^T))^{-1} \mathbf{X} \mathbf{\Lambda}^T \mathbf{U}^T \quad (4.53)$$

$$= (\mathbf{X} \mathbf{\Lambda}^T \mathbf{\Lambda} \mathbf{X}^T + \alpha^2 (\mathbf{X} \mathbf{M}^T \mathbf{M} \mathbf{X}^T))^{-1} \mathbf{X} \mathbf{\Lambda}^T \mathbf{U}^T \quad (4.54)$$

$$= (\mathbf{X} (\mathbf{\Lambda}^T \mathbf{\Lambda} + \alpha^2 \mathbf{M}^T \mathbf{M}) \mathbf{X}^T)^{-1} \mathbf{X} \mathbf{\Lambda}^T \mathbf{U}^T \quad (4.55)$$

$$= \mathbf{X}^{-T} (\mathbf{\Lambda}^T \mathbf{\Lambda} + \alpha^2 \mathbf{M}^T \mathbf{M})^{-1} \mathbf{\Lambda}^T \mathbf{U}^T. \quad (4.56)$$

The model resolution matrix is then

$$\mathbf{R}_m = \mathbf{G}^\# \mathbf{G} \quad (4.57)$$

$$= \mathbf{X}^{-T} (\mathbf{\Lambda}^T \mathbf{\Lambda} + \alpha^2 \mathbf{M}^T \mathbf{M})^{-1} \mathbf{\Lambda}^T \mathbf{U}^T \mathbf{U} \mathbf{\Lambda} \mathbf{X}^T \quad (4.58)$$

$$= \mathbf{X}^{-T} \mathbf{F} \mathbf{X}^T, \quad (4.59)$$

where

$$\mathbf{F} = (\mathbf{\Lambda}^T \mathbf{\Lambda} + \alpha^2 \mathbf{M}^T \mathbf{M})^{-1} \mathbf{\Lambda}^T \mathbf{\Lambda} \quad (4.60)$$

is a diagonal matrix of GSVD filter factors (4.50).

Example 4.5

To examine the resolution of the Tikhonov regularized inversions of [Example 4.4](#), we perform a spike test using (4.59). [Figure 4.17](#) shows the effect of multiplying $\mathbf{R}_m$ times a unit amplitude spike model (at depth 500 m) under first- and second-order Tikhonov regularization using α values of 122 and 2341, respectively. These curves can equivalently be conceptualized as rows/columns of the full resolution matrix at the index corresponding to 500 m. The spike test results indicate that these Tikhonov regularized solutions are smoothed versions of the spike model. Under first- or second-order regularization, the resolution of various model features will depend critically on how smooth or rough these features are in the true model. In [Figures 4.13](#) and [4.15](#), the higher-order solutions recover the true model better because the true model is smooth. Conversely, the spike model is not well recovered because of its rapid variation.

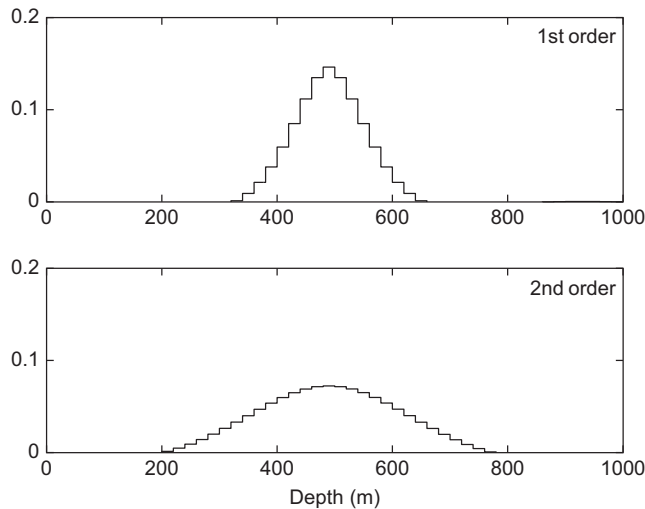


Figure 4.17 The model resolution matrix $\mathbf{R}_m$ multiplied times the spike model for the first- and second-order regularized solutions of [Example 4.4](#).

4.6. THE TGSVD METHOD

In the discussion of the SVD in Chapter 3, we examined the TSVD method of regularization, which rejects model space basis vectors associated with smaller singular values. Equivalently, this can be thought of as a damped SVD solution in which filter factors of one are used for basis vectors associated with larger singular values, and filter factors of zero are used for basis vectors associated with smaller singular values. This approach can be extended to the GSVD solution (4.49) to produce a **truncated generalized singular value decomposition** or **TGSVD** solution. In the TGSVD solution we simply assign filter factors (4.50) of one to the q largest generalized singular values terms in the sum to obtain

$$\mathbf{m}_{q,L} = \sum_{i=n-q+1}^n \frac{\mathbf{U}_{:,i+k}^T \mathbf{d}}{\lambda_i} \mathbf{Y}_{:,i} \quad (4.61)$$

Example 4.6

Applying the TGSVD method to the VSP problem, we find L-curve corners near $q = 8$ in the first-order case shown in Figure 4.18, and $q = 7$ in the second-order case shown in Figure 4.19. Examining the filter factors obtained for the corresponding Tikhonov solutions shown in Figure 4.16, we find that they decay precipitously with decreasing index near these locations. Figures 4.20 and 4.21 show the corresponding TGSVD solutions.

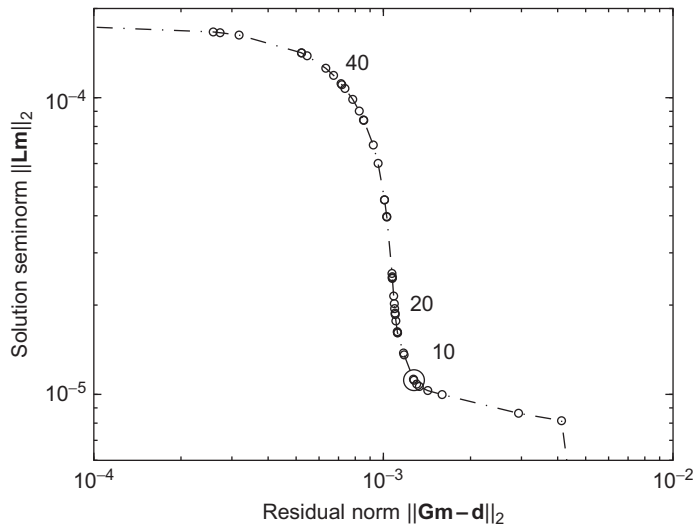


Figure 4.18 TGSVD L-curve for the VSP problem as a function of q for first-order regularization with the $q = 8$ solution indicated.

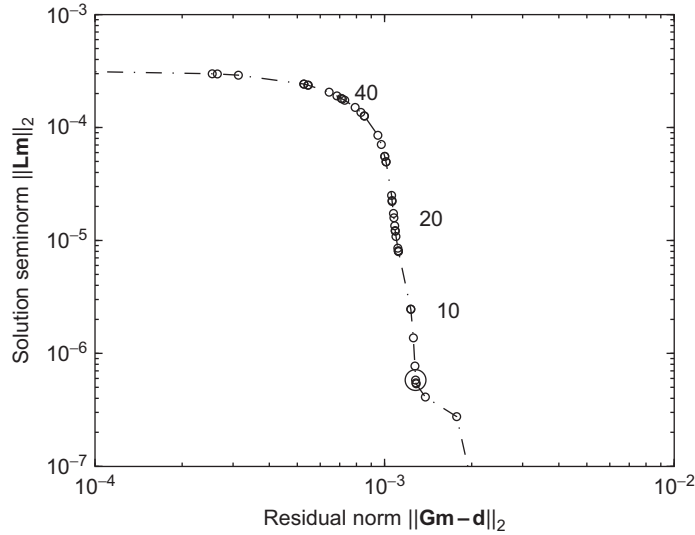


Figure 4.19 TGSVD L-curve for the VSP problem as a function of q for second-order regularization with the $q = 7$ solution indicated.

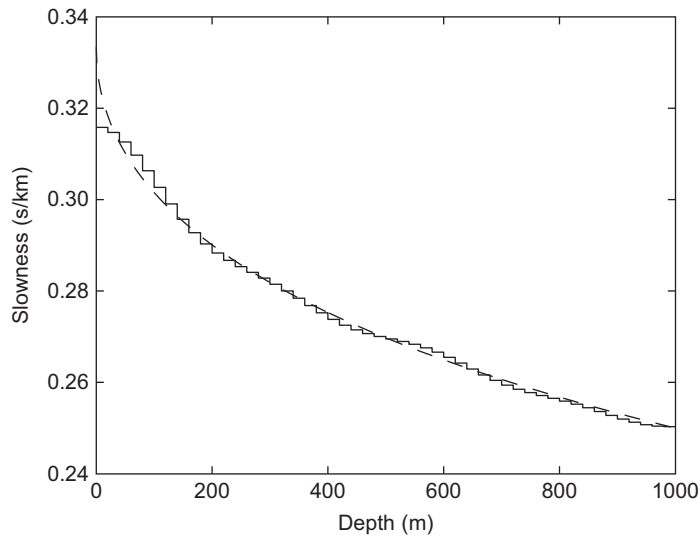


Figure 4.20 TGSVD solution of the VSP problem, $q = 8$, first-order regularization, shown in comparison with the true model.

The model recovery is comparable to that obtained with the Tikhonov method. The 2-norms of the difference between the first- and second-order solutions and the true model are approximately 1.0×10^{-2} s/km and 7.1×10^{-3} s/km, respectively, which are similar to the Tikhonov solutions in [Example 4.4](#).

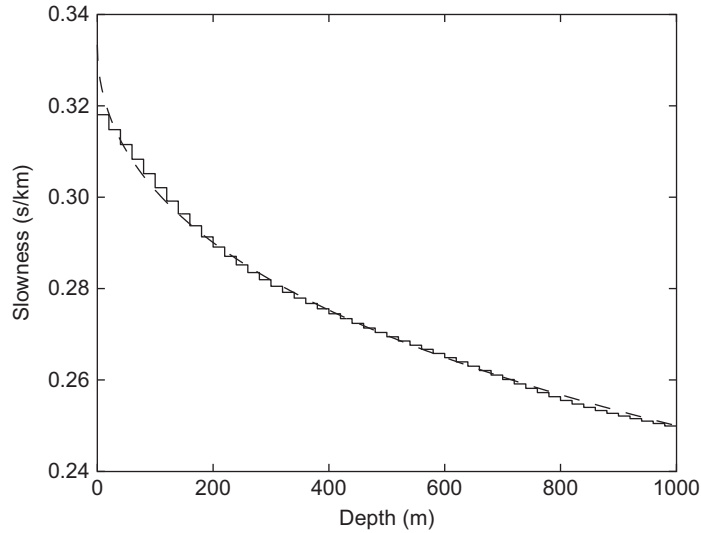


Figure 4.21 TGSVD solution of the VSP problem, $q = 7$, second-order regularization.

4.7. GENERALIZED CROSS-VALIDATION

Generalized cross-validation (GCV) is an alternative method for selecting a regularization parameter, α , that has a number of desirable statistical properties.

In ordinary or “leave-one-out” cross-validation, we consider the models that are obtained by leaving one of the m data points out of the fitting process. Consider the modified Tikhonov regularization problem in which we ignore a data point d_k ,

$$\min \sum_{i \neq k} ((\mathbf{G}\mathbf{m})_i - d_i)^2 + \alpha^2 \|\mathbf{L}\mathbf{m}\|_2^2 \quad (4.62)$$

Call the solution to this problem $\mathbf{m}_{\alpha,L}^{[k]}$, where the superscript indicates that d_k was left out of the computation. Ideally, the model $\mathbf{m}_{\alpha,L}^{[k]}$ would accurately predict the missing data value d_k . In the leave-one-out approach, we select the regularization parameter α so as to minimize the predictive errors for all k :

$$\min g(\alpha) = \frac{1}{m} \sum_{k=1}^m ((\mathbf{G}\mathbf{m}_{\alpha,L}^{[k]})_k - d_k)^2. \quad (4.63)$$

Unfortunately, computing $g(\alpha)$ involves solving m problems of the form (4.62). Generalized cross-validation is a way to speed up this computation.

First, let

$$\tilde{d}_i = \begin{cases} (\mathbf{Gm}_{\alpha,L}^{[k]})_k & i = k \\ d_i & i \neq k. \end{cases} \quad (4.64)$$

Note that because $(\mathbf{Gm}_{\alpha,L}^{[k]})_k = \tilde{d}_k$, $\mathbf{m}_{\alpha,L}^{[k]}$ also solves

$$\min((\mathbf{Gm})_k - \tilde{d}_k)^2 + \sum_{i \neq k} ((\mathbf{Gm})_i - \tilde{d}_i)^2 + \alpha^2 \|\mathbf{Lm}\|_2^2, \quad (4.65)$$

which is equivalent to

$$\min \|\mathbf{Gm} - \tilde{\mathbf{d}}\|_2^2 + \alpha^2 \|\mathbf{Lm}\|_2^2. \quad (4.66)$$

This result is known as the **leave-one-out lemma**. By this lemma,

$$\mathbf{m}_{\alpha,L}^{[k]} = \mathbf{G}^\# \tilde{\mathbf{d}}. \quad (4.67)$$

We will use (4.67) to eliminate $\mathbf{m}_{\alpha,L}^{[k]}$ from (4.63), because

$$\frac{(\mathbf{G}\mathbf{G}^\# \tilde{\mathbf{d}})_k - (\mathbf{G}\mathbf{G}^\# \mathbf{d})_k}{\tilde{d}_k - d_k} = (\mathbf{G}\mathbf{G}^\#)_{k,k}, \quad (4.68)$$

where $(\mathbf{G}\mathbf{G}^\#)_{k,k}$ are the diagonal elements of the data resolution matrix (4.21). Subtracting both sides of the equation from 1 gives

$$\frac{\tilde{d}_k - d_k - (\mathbf{G}\mathbf{G}^\# \tilde{\mathbf{d}})_k + (\mathbf{G}\mathbf{G}^\# \mathbf{d})_k}{\tilde{d}_k - d_k} = 1 - (\mathbf{G}\mathbf{G}^\#)_{k,k}. \quad (4.69)$$

Since $(\mathbf{G}\mathbf{G}^\# \mathbf{d})_k = (\mathbf{Gm}_{\alpha,L})_k$, $(\mathbf{G}\mathbf{G}^\# \tilde{\mathbf{d}})_k = \tilde{d}_k$, and $(\mathbf{Gm}_{\alpha,L}^{[k]})_k = \tilde{d}_k$, equation (4.69) simplifies to

$$\frac{(\mathbf{Gm}_{\alpha,L})_k - d_k}{(\mathbf{Gm}_{\alpha,L}^{[k]})_k - d_k} = 1 - (\mathbf{G}\mathbf{G}^\#)_{k,k}. \quad (4.70)$$

Rearranging this formula and substituting into (4.63), we obtain

$$g(\alpha) = \frac{1}{m} \sum_{k=1}^m \left(\frac{(\mathbf{Gm}_{\alpha,L})_k - d_k}{1 - (\mathbf{G}\mathbf{G}^\#)_{k,k}} \right)^2. \quad (4.71)$$

We can simplify the formula further by replacing the $(\mathbf{G}\mathbf{G}^\sharp)_{k,k}$ with the average value

$$(\mathbf{G}\mathbf{G}^\sharp)_{k,k} \approx \frac{1}{m} \text{Tr}(\mathbf{G}\mathbf{G}^\sharp), \quad (4.72)$$

which gives an expression for the (4.63) that can be evaluated as a function of α :

$$g(\alpha) \approx \frac{1}{m} \sum_{k=1}^m \frac{((\mathbf{G}\mathbf{m}_{\alpha,L})_k - d_k)^2}{\left(\frac{1}{m}(m - \text{Tr}(\mathbf{G}\mathbf{G}^\sharp))\right)^2} \quad (4.73)$$

$$= \frac{m \|\mathbf{G}\mathbf{m}_{\alpha,L} - \mathbf{d}\|_2^2}{\text{Tr}(\mathbf{I} - \mathbf{G}\mathbf{G}^\sharp)^2}. \quad (4.74)$$

It can be shown that under reasonable assumptions regarding the noise and smoothness of $\mathbf{m}_{\text{true}}$, the value of α that minimizes (4.74) approaches the value that minimizes $E[\mathbf{G}\mathbf{m}_{\alpha,L} - \mathbf{d}_{\text{true}}]$ as the number of data points m goes to infinity, and that under the same assumptions, $E[\|\mathbf{m}_{\text{true}} - \mathbf{m}_{\alpha,L}\|_2]$ goes to 0 as m goes to infinity [36, 170]. In practice, the size of the data set is fixed in advance, so the limit is not directly applicable. However, these results provide a theoretical justification for using the GCV method to select the Tikhonov regularization parameter.

Example 4.7

Figures 4.22 and 4.23 show $g(\alpha)$ for the VSP test problem, using first- and second-order Tikhonov regularization, respectively. Respective GCV function (4.74) minima occur

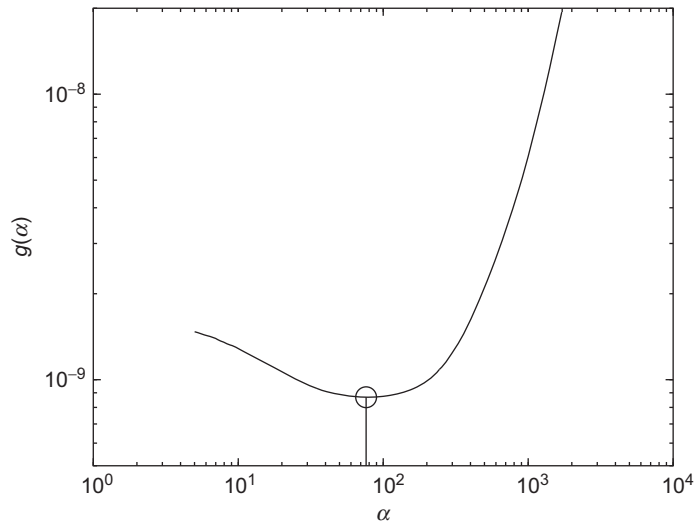


Figure 4.22 GCV curve for the VSP problem and its minimum, first-order regularization.

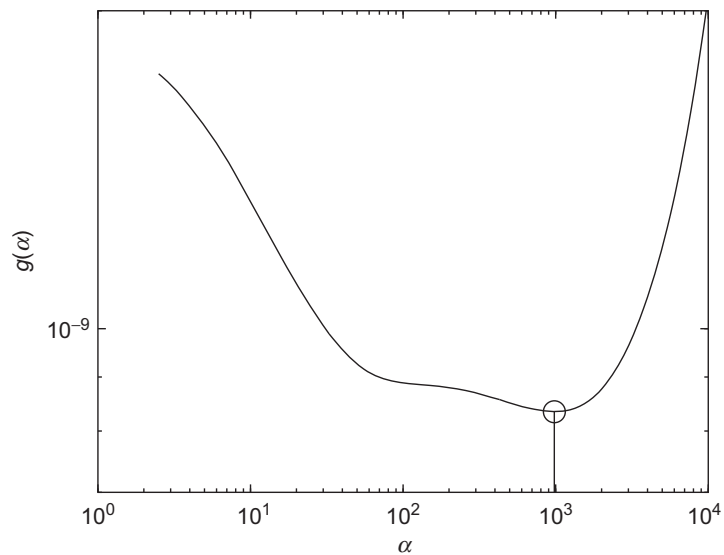


Figure 4.23 GCV curve for the VSP problem and its minimum, second-order regularization.

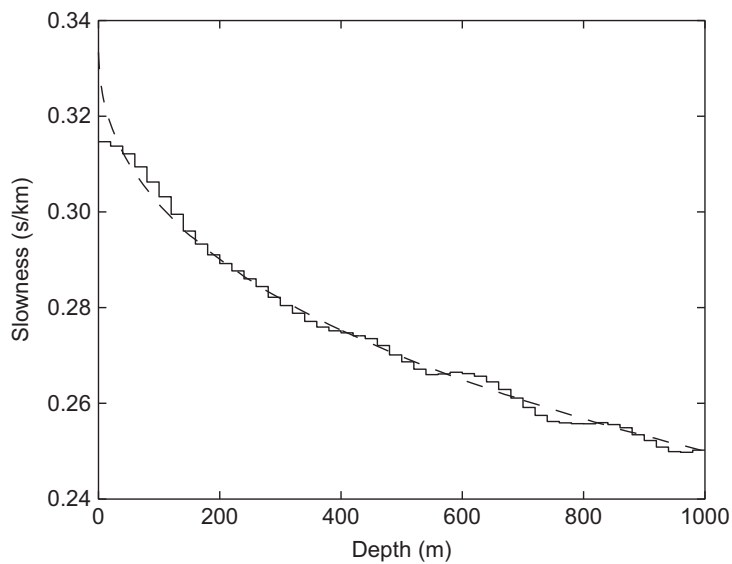


Figure 4.24 GCV solution for the VSP problem, first-order, $\alpha = 76.3$, shown in comparison with the true model.

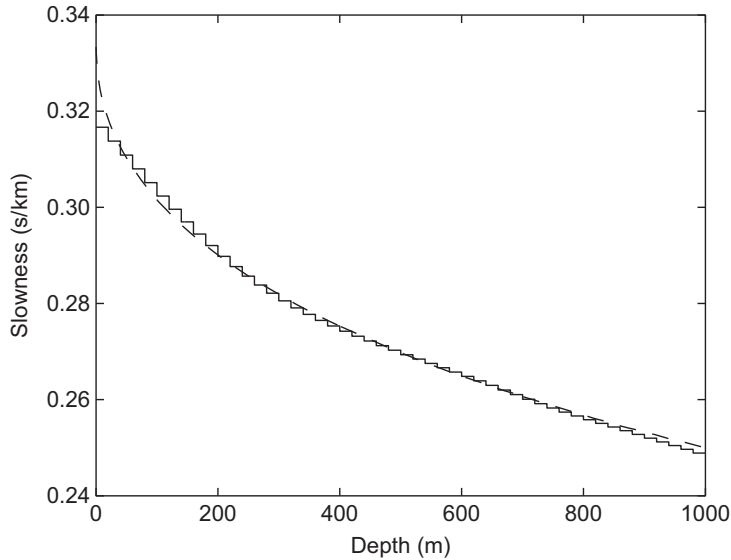


Figure 4.25 GCV solution for the VSP problem, second-order, $\alpha = 981$, shown in comparison with the true model.

near $\alpha = 76.3$ and $\alpha = 981$, which are somewhat smaller than the α values estimated previously using the L-curve (Example 4.4). The corresponding models (Figures 4.24 and 4.25) thus have somewhat larger seminorms.

4.8. ERROR BOUNDS

We next present two theoretical results that help to address the accuracy of Tikhonov regularization solutions. We will present these results in a simplified form, covering only zeroth-order Tikhonov regularization.

The first question is whether for a particular value of the regularization parameter, α , we can establish a bound on the sensitivity of the regularized solution to the noise in the observed data $\mathbf{d}$ and/or errors in the system matrix $\mathbf{G}$. This would provide a sort of condition number for the inverse problem. Note that this does not tell us how far the regularized solution is from the true model, since Tikhonov regularization has introduced a bias in the solution. Under Tikhonov regularization with a nonzero α , we would not obtain the true model even if the noise was 0.

The following theorem gives a bound for zeroth-order Tikhonov regularization [65]. A slightly more complicated formula is available for higher-order Tikhonov regularization [65].

Theorem 4.1 *Suppose that the problems*

$$\min \|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{m}\|_2^2 \quad (4.75)$$

and

$$\min \|\bar{\mathbf{G}}\mathbf{m} - \bar{\mathbf{d}}\|_2^2 + \alpha^2 \|\mathbf{m}\|_2^2 \quad (4.76)$$

are solved to obtain $\mathbf{m}_\alpha$ and $\bar{\mathbf{m}}_\alpha$. Then

$$\frac{\|\mathbf{m}_\alpha - \bar{\mathbf{m}}_\alpha\|_2}{\|\mathbf{m}_\alpha\|_2} \leq \frac{\bar{\kappa}_\alpha}{1 - \epsilon \bar{\kappa}_\alpha} \left(2\epsilon + \frac{\|\mathbf{e}\|_2}{\|\mathbf{d}_\alpha\|_2} + \epsilon \bar{\kappa}_\alpha \frac{\|\mathbf{r}_\alpha\|_2}{\|\mathbf{d}_\alpha\|_2} \right) \quad (4.77)$$

where

$$\bar{\kappa}_\alpha = \frac{\|\mathbf{G}\|_2}{\alpha} \quad (4.78)$$

$$\mathbf{E} = \mathbf{G} - \bar{\mathbf{G}} \quad (4.79)$$

$$\mathbf{e} = \mathbf{d} - \bar{\mathbf{d}} \quad (4.80)$$

$$\epsilon = \frac{\|\mathbf{E}\|_2}{\|\mathbf{G}\|_2} \quad (4.81)$$

$$\mathbf{d}_\alpha = \mathbf{G}\mathbf{m}_\alpha \quad (4.82)$$

and

$$\mathbf{r}_\alpha = \mathbf{d} - \mathbf{d}_\alpha. \quad (4.83)$$

In the particular case when $\mathbf{G} = \bar{\mathbf{G}}$, and the only difference between the two problems is $\mathbf{e} = \mathbf{d} - \bar{\mathbf{d}}$, the inequality becomes even simpler

$$\frac{\|\mathbf{m}_\alpha - \bar{\mathbf{m}}_\alpha\|_2}{\|\mathbf{m}_\alpha\|_2} \leq \bar{\kappa}_\alpha \frac{\|\mathbf{e}\|_2}{\|\mathbf{d}_\alpha\|_2}. \quad (4.84)$$

The condition number $\bar{\kappa}_\alpha$ is inversely proportional to α . Thus increasing α will decrease the sensitivity of the solution to perturbations in the data. Of course, increasing α also increases the error in the solution due to regularization bias and decreases resolution.

The second question is whether we can establish any sort of bound on the norm of the difference between the regularized solution and the true model. This bound would incorporate both sensitivity to noise and the bias introduced by Tikhonov regularization. Such a bound must of course depend on the magnitude of the noise in the data. It must also depend on the particular regularization parameter chosen. Tikhonov developed a beautiful theorem that addresses this question in the context of inverse problems involving IFKs [161]. More recently, Neumaier has developed a version of Tikhonov's theorem that can be applied directly to discretized problems [116].

Recall that in a discrete ill-posed linear inverse problem, the matrix $\mathbf{G}$ commonly has a smoothing effect, in that when we multiply $\mathbf{G}\mathbf{m}$, the result is smoother than $\mathbf{m}$. Similarly, if we multiply $\mathbf{G}\mathbf{m}$ by $\mathbf{G}^T$, the result will be even smoother than $\mathbf{G}\mathbf{m}$. This smoothing in such problems is a consequence of the fact that the singular vectors corresponding to the larger singular values of $\mathbf{G}$ are smooth. Note, however, that this is not a universal property of all matrices; for example, if $\mathbf{G}$ is a matrix that approximates the differentiation operator, then $\mathbf{G}\mathbf{m}$ will be rougher than $\mathbf{m}$.

For discrete ill-posed problems, models in the range of $\mathbf{G}^T$ will form a relatively smooth subspace of all possible models in R^n . Models in this subspace can be written using the range of $\mathbf{G}^T$ as a basis as $\mathbf{m} = \mathbf{G}^T \mathbf{w}$, using the coefficients $\mathbf{w}$. Furthermore, models in the range of $\mathbf{G}^T \mathbf{G}$ form a subspace of $R(\mathbf{G}^T)$, since any model in $R(\mathbf{G}^T \mathbf{G})$ can be written as $\mathbf{m} = \mathbf{G}^T (\mathbf{G}\mathbf{w})$, which is a linear combination of columns of $\mathbf{G}^T$. Because of the smoothing effect of $\mathbf{G}$ and $\mathbf{G}^T$, we would expect these models to be even smoother than the models in $R(\mathbf{G}^T)$. We could construct smaller subspaces of R^n that contain even smoother models, but it turns out that with zeroth-order Tikhonov regularization these are the only subspaces of interest.

There is another way to see that models in $R(\mathbf{G}^T)$ will be relatively smooth. Recall that the vectors $\mathbf{V}_{\cdot,1}, \mathbf{V}_{\cdot,2}, \dots, \mathbf{V}_{\cdot,p}$ from the SVD of $\mathbf{G}$ form an orthonormal basis for $R(\mathbf{G}^T)$. For discrete ill-posed problems, we know from Chapter 3 that these basis vectors will be relatively smooth, so linear combinations of these vectors in $R(\mathbf{G}^T)$ should be smooth.

The following theorem gives a bound on the total error including bias due to regularization and error due to noise in the data for zeroth-order Tikhonov regularization [116].

Theorem 4.2 *Suppose that we use zeroth-order Tikhonov regularization with regularization parameter α to solve $\mathbf{G}\mathbf{m} = \mathbf{d}$, and that $\mathbf{m}_{\text{true}}$ can be expressed as one of the following distinct cases, for some $\mathbf{w}$, and as parameterized by p ,*

$$\mathbf{m}_{\text{true}} = \begin{cases} \mathbf{G}^T \mathbf{w} & p = 1 \\ \mathbf{G}^T \mathbf{G} \mathbf{w} & p = 2 \end{cases} \quad (4.85)$$

and that

$$\|\mathbf{G}\mathbf{m}_{\text{true}} - \mathbf{d}\|_2 \leq \Delta \|\mathbf{w}\|_2 \quad (4.86)$$

for some $\Delta > 0$. Then

$$\|\mathbf{m}_{\text{true}} - \mathbf{G}^\# \mathbf{d}\|_2 \leq \left(\frac{\Delta}{2\alpha} + \gamma \alpha^p \right) \|\mathbf{w}\|_2, \quad (4.87)$$

where

$$\gamma = \begin{cases} 1/2 & p = 1 \\ 1 & p = 2 \end{cases}. \quad (4.88)$$

Furthermore, if we begin with the bound

$$\|\mathbf{G}\mathbf{m}_{\text{true}} - \mathbf{d}\|_2 \leq \delta, \quad (4.89)$$

we can let

$$\Delta = \frac{\delta}{\|\mathbf{w}\|_2}. \quad (4.90)$$

Under this condition the optimal value of α is

$$\hat{\alpha} = \left(\frac{\Delta}{2\gamma p} \right)^{\frac{1}{p+1}} = O(\Delta^{\frac{1}{p+1}}). \quad (4.91)$$

With this choice of α ,

$$\Delta = 2\gamma p \hat{\alpha}^{p+1} \quad (4.92)$$

and the error bound simplifies to

$$\|\mathbf{m}_{\text{true}} - \mathbf{G}_{\hat{\alpha}}^{\dagger} \mathbf{d}\|_2 \leq \gamma(p+1) \hat{\alpha}^p \|\mathbf{w}\|_2 = O(\Delta^{\frac{p}{p+1}}). \quad (4.93)$$

Theorem 4.2 tells us that the error in the Tikhonov regularization solution depends on both the noise level δ and on the regularization parameter α . For larger values of α , the error due to regularization in recovering the true model will be dominant. For smaller values of α , the error due to noise in the data will be dominant. We seek the optimal value of α that best balances these effects. Using the optimal α , (4.93) indicates that we can obtain an error bound of $O(\Delta^{2/3})$ if $p = 2$, and an error bound of $O(\Delta^{1/2})$ if $p = 1$.

Of course, the above result can only be applied when our true model lives in the restricted subspace of $R(\mathbf{G}^T)$. In practice, even if the model does lie in $R(\mathbf{G}^T)$, the vector $\mathbf{w}$ could have a very large norm, making the bound uninformative.

Thus, applying this theorem in a quantitative fashion is typically impractical. However, the theorem does provide some useful insight into the ability of Tikhonov regularization to recover a true model. The first point is that the accuracy of the regularized solution depends very much on the smoothness of the true model. If $\mathbf{m}_{\text{true}}$ is not smooth, then Tikhonov regularization simply will not give an accurate solution. Furthermore, if the model $\mathbf{m}_{\text{true}}$ is smooth, then we can hope for an error in the Tikhonov

regularized solution which is $O(\delta^{1/2})$ or $O(\delta^{2/3})$. Another way of saying this is that we can hope *at best* for an answer with about two thirds as many correct significant digits as the data.

Example 4.8

Recall the Shaw problem previously considered in Examples 4.1 and 4.3. Because $\mathbf{G}^T$ is a nonsingular matrix, the spike model should lie in $R(\mathbf{G}^T)$. However, $\mathbf{G}^T$ is numerically singular due to the ill-posedness of the problem, and the spike model thus lies outside of the effective range of $\mathbf{G}^T$. Any unregularized attempt to find $\mathbf{w}$ produces a meaningless answer due to numerical instability. Because the spike model does not lie in $R(\mathbf{G}^T)$, Theorem 4.2 does not apply.

Figure 4.26 shows a smooth model that does lie in the range of $\mathbf{G}^T$. For this model we constructed a synthetic data set with noise as before at $\delta = 4.47 \times 10^{-6}$. Equation (4.93) suggests using $\alpha = 8.0 \times 10^{-4}$. The resulting error bound is 8.0×10^{-4} , while the actual norm of the model error is 6.6×10^{-4} . Here the data were accurate to roughly six digits, while the solution was accurate to roughly four digits. Figure 4.27 shows the reconstruction of the model with $N(0, (1.0 \times 10^{-6})^2)$ noise added to the data vector, showing that the solution is well recovered. This example once again demonstrates the importance of smoothness in the true model in determining how accurately it can be reconstructed with Tikhonov regularization.

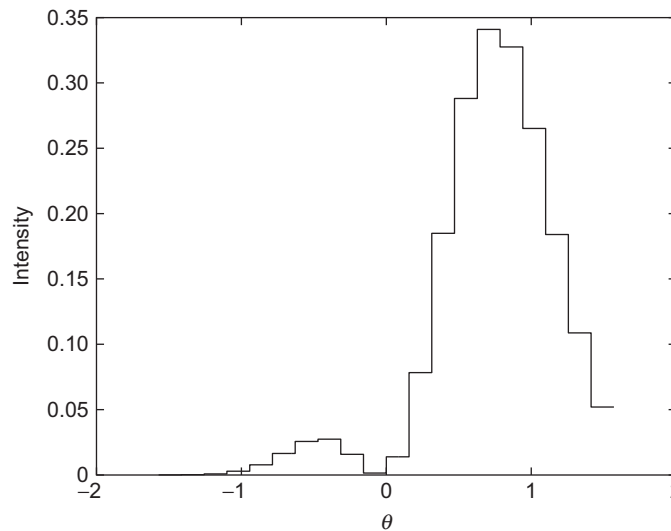


Figure 4.26 A smooth model in $R(\mathbf{G}^T)$.

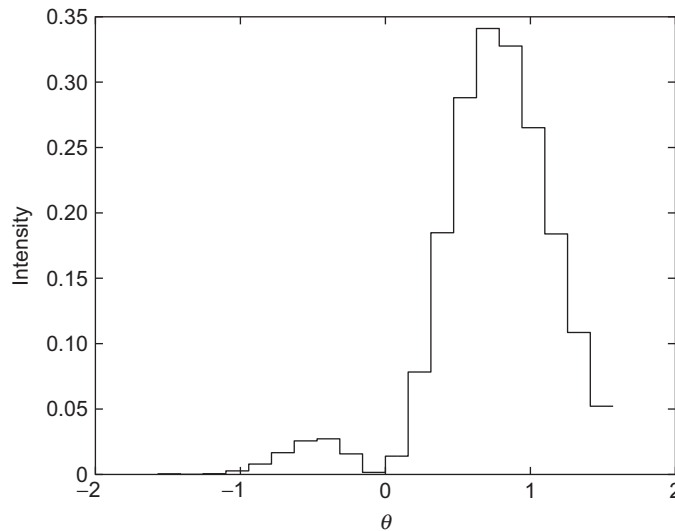


Figure 4.27 Reconstruction of the smooth model with $N(0, (1.0 \times 10^{-6})^2)$ noise.

4.9. EXERCISES

1. Use the method of Lagrange multipliers (Appendix C) to derive the damped least squares problem (4.4) from the discrepancy principle problem (4.2), and demonstrate that (4.4) can be written as (4.5).
2. Consider the integral equation and data set from Problem 3.5. You can find a copy of this data set in the file **ifk.mat**.
 - a. Discretize the problem using simple collocation.
 - b. Using the data supplied, and assuming that the numbers are accurate to four significant figures, determine a reasonable bound δ for the misfit.
 - c. Use zeroth-order Tikhonov regularization to solve the problem. Use GCV, the discrepancy principle, and the L-curve criterion to pick the regularization parameter.
 - d. Use first-order Tikhonov regularization to solve the problem. Use GCV, the discrepancy principle, and the L-curve criterion to pick the regularization parameter.
 - e. Use second-order Tikhonov regularization to solve the problem. Use GCV, the discrepancy principle, and the L-curve criterion to pick the regularization parameter.
 - f. Analyze the resolution of your solutions. Are the features you see in your inverse solutions unambiguously real? Interpret your results. Describe the size and location of any significant features in the solution.





3. Consider the following problem in **cross-well tomography**. Two vertical wells are located 1600 m apart. A seismic source is inserted in one well at depths of 50, 150, . . . , 1550 m. A string of receivers is inserted in the other well at depths of 50 m, 150 m, . . . , 1550 m. See Figure 4.28. For each source-receiver pair, a travel time is recorded, with a measurement standard deviation of 0.5 ms. There are 256 ray paths and 256 corresponding data points. We wish to determine the velocity structure in the two-dimensional plane between the two wells.

Discretizing the problem into a 16 by 16 grid of 100 meter by 100 meter blocks gives 256 model parameters. The **G** matrix and noisy data, **d**, for this problem (assuming straight ray paths) are in the file **crosswell.mat**. The order of parameter indexing from the slowness grid to the model vector is row by row (e.g., Example 1.12).

- Use the TSVD to solve this inverse problem using an L-curve. Plot the result.
- Use zeroth-order Tikhonov regularization to solve this problem and plot your solution. Explain why it is difficult to use the discrepancy principle to select the regularization parameter. Use the L-curve criterion to select your regularization parameter. Plot the L-curve as well as your solution.
- Use second-order Tikhonov regularization to solve this problem and plot your solution. Because this is a two-dimensional problem, you will need to implement a finite-difference approximation to the Laplacian (second derivative in the

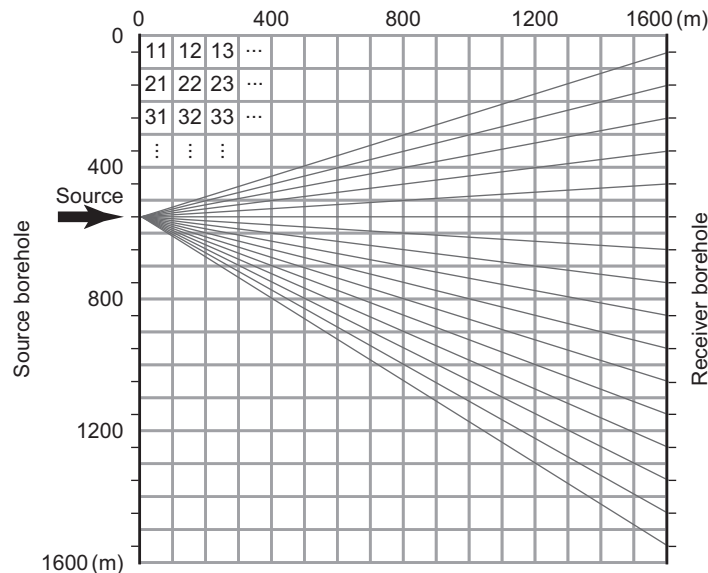


Figure 4.28 Cross-well tomography problem, showing block discretization, block numbering convention, and one set of straight source-receiver ray paths.

horizontal direction plus the second derivative in the vertical direction) in the roughening matrix. The $\mathbf{L}$ matrix can be generated using the following MATLAB code:

```
L=zeros(14*14,256);
k=1;
for i=2:15,
    for j=2:15,
        M=zeros(16,16);
        M(i,j)=-4;
        M(i,j+1)=1;
        M(i,j-1)=1;
        M(i+1,j)=1;
        M(i-1,j)=1;
        L(k,:)=reshape(M,256,1)';
        k=k+1;
    end
end
```

What, if any, problems did you have in using the L-curve criterion on this problem? Plot the L-curve as well as your solution.

- d. Discuss your results. If vertical bands appeared in some of your solutions, can you explain why?
4. Apply second-order Tikhonov regularization to solve the problem of finding a smooth curve that approximately fits a set of noisy data points. Write a MATLAB program to find a function specified at the 191 points $x = 1, 1.1, 1.2, \dots, 20$ that approximately fits the 20 data points specified at $x = 1, 2, 3, \dots, 20$ given in the file **interpdata.mat**. Construct an appropriate 20 by 191 $\mathbf{G}$ matrix, and use the library function **get1rough** to obtain the necessary second-order roughening matrix, $\mathbf{L}$. Produce solutions for regularization parameter values of $\alpha = 0.2, 0.4, 0.6, \dots, 10$ and show the tradeoff curve between 2-norm data misfit and model seminorm on a linear-linear plot. If the data noise is independent and normally distributed with a standard deviation of 0.3, use the discrepancy principle to find and plot an optimal interpolated curve along with the data points. What is the χ^2 value of this solution? Is it reasonable?
5. In some situations it is appropriate to bias the regularized solution toward a particular model $\mathbf{m}_0$. In this case, we would solve

$$\min \|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{L}(\mathbf{m} - \mathbf{m}_0)\|_2^2. \quad (4.94)$$

Write this as an ordinary linear least squares problem. What are the normal equations? Can you find a solution for this problem using the GSVD?

4.10. NOTES AND FURTHER READING

Hansen's book [65] is a very complete reference on the linear algebra of Tikhonov regularization. Arnold Neumaier's tutorial [116] is also a very useful reference. Two other surveys of Tikhonov regularization are [44, 45]. Vogel [169] includes an extensive discussion of methods for selecting the regularization parameter. Hansen's MATLAB Regularization Tools [64] is a collection of software for performing regularization within MATLAB.

The GSVD was first defined by Van Loan [101]. References on the GSVD and algorithms for computing the GSVD include [3, 53, 62, 65].

Selecting the regularization parameter is an important problem in both theory and practice. Much of the literature on functional analytic approaches assumes that the noise level is known. When the noise level is known, the discrepancy principle provides a scheme for selecting the regularization parameter for ill-posed problems that is convergent in the sense that, in the limit as the noise level goes to zero, the regularized solution goes to $\mathbf{m}_{\text{true}}$ [45].

In practice, the noise level is often unknown, so there has been a great deal of interest in schemes for selecting the regularization parameter without its prior knowledge. The two most popular approaches are the L-curve method and GCV. The L-curve method was introduced by Hansen [63, 65]. GCV was introduced by Craven and Wahba [36, 170]. The formula for GCV given here is very expensive to compute for large problems. A GCV algorithm for large-scale problems is given by Golub and von Matt [54]. Vogel has shown that the L-curve method can fail to converge [168]. It can be shown that no scheme that depends only on noisy data without knowledge of the noise level can be convergent in the limit as the noise level goes to zero [45].

Within statistics, poorly conditioned linear regression problems are said to suffer from "multicollinearity." A method called "ridge regression," which is identical to Tikhonov regularization, is often used to deal with such problems [40]. Statisticians also use a method called "principal components regression" (PCR), which is identical to the TSVD method [113].